Matthew Faw

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Research Interests

Stochastic Optimization, Online Decision-Making, Reinforcement Learning and Control, Time-series Forecasting

Education

Georgia Institute of Technology **Atlanta, GA**
ARC Postdoctoral Fellow (Focus: Model-free/online control, Online resource allocation) *2024–present*
Mentors: Siva Theja Maguluri, Sahil Singla

The University of Texas at Austin **Austin, TX**
Ph.D. Machine Learning (Thesis: Adaptive Algorithms for Stochastic Optimization and Bandit Learning), *2018–2024*
Advisors: Sanjay Shakkottai, Constantine Caramanis.

Duke University **Durham, NC**
B.S.E. Electrical & Computer Engineering, B.S. Computer Science, A.B. Math (cum laude), *2013–2017*
Advisors: Nick Buchler, Richard Fair, Benjamin C. Lee

Industry Experience

Google Research **Remote**
Visiting Researcher *March 2025–September 2025*
Designed, trained, and evaluated novel discrete tokenization approaches to time-series forecasting with transformers.

Google Research **Mountain View, CA**
Ph.D. Student Researcher *June-September 2024*
Designed, trained, and evaluated an in-context learning approach for time-series forecasting using a decoder-only transformer architecture.

Verato **McLean, VA**
Software Engineer (Full-time) *June 2017-July 2018*
Co-designed and built a custom CI system using Kubernetes. Created software for entity search, updates, and analysis.

MathWorks **Natick, MA**
Software Engineering Intern, Stateflow Semantics *May-August 2016*
Co-designed + implemented proof-of-concept architectural change to code generation process to improve extensibility and performance.

Selected Publications (Full List on Google Scholar)

In Submission

2026: "Online Allocation Using Few Samples", **F**, S. Singla, Y. Wang (alphabetical order, *Georgia Tech*)
Developed a new algorithm for online resource allocation, load balancing, and packing/covering LPs, obtaining near-optimal results (in resource allocation) and significantly improved results (in the other settings).

2026: "Online Control with Multiple Sensors", **F**, S. T. Maguluri (*Georgia Tech*)
Proposed a generalization of the online control problem, and proved near-optimal regret upper-bounds for a novel algorithm.

Conference Papers

ICML 2025: "In-Context Fine-Tuning for Time-Series Foundation Models", **F**, R. Sen, Y. Zhou, A. Das (*Georgia Tech*, *Google Research*)
Empirical evaluation of a new approach to time-series forecasting using in-context learning and a decoder-only transformer model.

ICML 2025: "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback", **F**, C. Caramanis, J. Hoffmann (*Georgia Tech*, *UT Austin*, *Google DeepMind*)
Algorithms and new lower bounds for regret under an affinity-biased feedback model (preliminary results appeared at NeurIPS 2023 workshop: Algorithmic Fairness through the Lens of Time)

COLT 2023: "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD", **F**$^=$, L. Rout$^=$, C. Caramanis, S. Shakkottai (*UT Austin*)
First algorithm which converges at order-optimal rate under a generalized smoothness assumption in standard noise regime.

COLT 2022: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance", **F**$^=$, I. Tziotis$^=$, C. Caramanis, A. Mokhtari, S. Shakkottai, R. Ward (*UT Austin*)

Resolved an open problem on convergence of AdaGrad-Norm for smooth stochastic non-convex optimization.

SIGMETRICS 2022/POMACS: "Learning To Maximize Welfare with a Reusable Resource", **F**$^{=}$, O. Papadigenopoulos$^{=}$, C. Caramanis, S. Shakkottai (*UT Austin*)
Optimal prophet inequalities, learning variants, and lower bounds for prophet inequalities with dynamic constraints.

SODA 2022, Theory of Computing 2026: "Single-Sample Prophet Inequalities via Greedy-Ordered Selection", C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order, *UT Austin*, *Google Switzerland*, *Sapienza University of Rome*, *Drexel*)
Improved single-sample prophet inequalities for nearly all combinatorial settings considered in prior work.

NeurIPS 2020: "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions", **F**, R. Sen, K. Shanmugam, C. Caramanis, S. Shakkottai (*UT Austin*, *IBM Research*)
Optimistic bandit tree-search for multi-source domain adaptation.

Journal Papers

Theory of Computing 2026: "Single-Sample Prophet Inequalities via Greedy-Ordered Selection", C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order, *UT Austin*, *Google Switzerland*, *Sapienza University of Rome*, *Drexel*)
Journal version of SODA'22

POMACS 2022: "Learning to Maximize Welfare with a Reusable Resource", **F**$^{=}$, O. Papadigenopoulos$^{=}$, C. Caramanis, S. Shakkottai (*UT Austin*)
Journal version of SIGMETRICS'22

TOCS 2017: "Computational Sprinting: Architecture, Dynamics, and Strategies", S. Zahedi, S. Fan, **F**, E. Cole, B. Lee (*Duke University*)
Evaluated performance and economic viability of several strategies for system-level computational sprinting for Spark applications.

Awards + Honors

2023: Dr. Brooks Carlton Fowler Endowed Presidential Graduate Fellowship in ECE, 2023-2024 academic year

2022: Top 10% reviewer for NeurIPS'22 and AISTATS'22, Highlighted reviewer for ICLR 2022

2020: NXP Foundation Fellowship, 2020-2021 academic year

2017: Cum Laude Graduation Honors, Duke University

2016: Member, Tau Beta Pi and Eta Kappa Nu Honor Societies, Duke University

2014: Gold medal, International Genetically Engineered Machine Competition

Talks and Poster Presentations

Invited Talks

October 2025: INFORMS Job Market Showcase, Atlanta, GA: "Optimal Control with Multiple Sensors"

July 2025: INFORMS APS Conference, Atlanta, GA: "Fundamental Limits of Regret Minimization in Stochastic Bandits with Evolving Biased Feedback"

February 2025: Google Research, (Virtual): "Order-Optimal Convergence Rates with Adaptive SGD"

October 2024: INFORMS Annual Meeting, Seattle, WA: "Order-Optimal Convergence Rates with Adaptive SGD"

March 2024: Georgia Tech ARC Colloquium, Atlanta, GA: "The Power of Adaptivity in SGD"

Talks

April 2026: Applied Probability Group Meeting, Atlanta, GA: "Online Control with Multiple Sensors"

July 2023: COLT 2023, Bangalore, India: "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

April 2023: IFML Workshop, UW "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

July 2022: COLT 2022, London, UK: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

June 2022: SIGMETRICS 2022, IIT Bombay, Mumbai, IN: "Learning To Maximize Welfare with a Reusable Resource"

April 2022: Machine Learning Lab Research Symposium, UT Austin: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

January 2022: SODA 2022, Virtual: "Single Sample Prophet Inequalities via Greedy-Ordered Selection"

Poster Presentations

July 2025: ICML 2025, Vancouver, Canada, "In-Context Fine-Tuning for Time-Series Foundation Models" and "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"

May 2024: IFML NSF Site Visit Poster Session, UT Austin, "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

December 2023: NeurIPS 2023 Workshop: Algorithmic Fairness through the Lens of Time, New Orleans, LA, "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"

October 2022: Joint IFML/Data-Driven Decision Processes Workshop, Simons Institute, UC Berkeley, "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

December 2020: NeurIPS 2020, Virtual, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"

November 2019: Texas Wireless Summit, UT Austin, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"

Teaching Experience

Georgia Tech
ISyE 3770 Statistics and Applications ***Instructor*** — *Spring 2026*
Served as the instructor for introductory probability and statistics course. Designed and delivered lectures, office hours, homework, and exams for 60 undergraduate students. Managed two undergraduate TAs.

UT Austin
EE 460J, Data Science Lab TA — *Fall 2018, Spring 2019*
Led lab sessions, and graded homeworks and exams

Duke University
CS 308, Software Design and Implementation TA — *Spring 2017*
Personally mentored 3 undergraduate CS students, conducted code reviews, advised and evaluated design and implementation of 3 software projects.

Duke University
ECE 280, Signals & Systems TA — *Fall 2015*
Held office hours and graded assignments for 70 undergraduate students.

Duke University
Synthetic Biology House Course Co-Designer/Instructor — *Spring 2015*
Co-designed and taught the first-offered Duke course on synthetic biology to 10 undergraduates.

Conference Reviewing

2021-Present: AISTATS, ALT, ICLR, ICML, JMLR, NeurIPS

Technical Skills

Programming: Python, Java, C/C++, JavaScript

Infrastructure: Kubernetes, AWS, Google Cloud, Mongo, Solr

References

Sanjay Shakkottai (Ph.D. Advisor) sanjay.shakkottai@utexas.edu
Constantine Caramanis (Ph.D. Advisor) constantine@utexas.edu
Siva Theja Maguluri (Postdoc Mentor) siva.theja@gatech.edu